

Simple Linear Regression

February 28, 2024

Linear Regression

```
[1]: #STEPS
      #1) Importing Libraries
      #2) Identify a dataset (Exported dataset (.csv, .xlsx), just an array (defined ↵
            ↵dataset))
      #3) Create a model
      #4) Getting Results/ model evaluation
      #5) Prediction
```

```
[9]: #1) Importing Libraries
      import numpy as np
      from sklearn.linear_model import LinearRegression
```

```
[3]: #2) Identify a dataset (Imported dataset (.csv, .xlsx), just an array (defined ↵
            ↵dataset))
      #a) Array of data
      x = np.array([5, 10, 15, 20, 25]).reshape((-1,1))
      y = np.array([5, 20, 28, 33, 52])
```

```
[4]: x
```

```
[4]: array([[ 5],
            [10],
            [15],
            [20],
            [25]])
```

```
[5]: y
```

```
[5]: array([ 5, 20, 28, 33, 52])
```

```
[6]: #Import a dataset from an external source
      #1) Import pandas as pd
      #2) data_name = pd.read_csv("Data file path")
      #3) Read/ display the data, print bteh assigned name. (data_name)
```

```
[10]: #3) Create a model
model = LinearRegression()

[11]: #Model fitting, Matching the data we have with the already developed/ called
      ↪model
model.fit(x,y)

[11]: LinearRegression()

[12]: #The above two steps can be joined as
      #model = LinearRegression().fit(x,y)

[13]: #4) Getting Results/ model evaluation
results = model.score(x,y)
results

[13]: 0.9595206168286959

[14]: #Intercepts
model.intercept_

[14]: -4.5

[15]: #Slope/ gradient
model.coef_

[15]: array([2.14])

[16]: #Regression line  $y = 2.14x - 4.5$ 

[18]: #Make a prediction
y_pred = model.predict(x)

[19]: y_pred

[19]: array([ 6.2, 16.9, 27.6, 38.3, 49. ])

[ ]:
```